

Functional Requirement Document (FRD)

Clans Machina Residential Solar Savings Calculator

Objective

Develop an intelligent Solar Savings Calculator that enables website visitors to estimate their ideal rooftop solar system, project investment, subsidy eligibility, EMI, savings, environmental impact, and long-term financial benefits. The calculator should also generate a professional PDF report and recommend the most suitable solar system using predefined business logic

User Inputs

1. State

- Odisha (Current Version)
- Other States (Future Expansion)

2. District

Dropdown list (for future DISCOM-wise integration)

3. Average Monthly Electricity Bill (₹)

Numeric input

4. Available Shadow-Free Roof Area (Sq. Ft.)

Numeric input

5. Payment Option

- Cash / Self Finance
- Bank Loan

6. Loan Tenure (Visible only for Bank Loan)

- 3 Years
- 5 Years

- 7 Years
 - 10 Years
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Business Logic

Step 1 – Calculate Monthly Electricity Consumption

Electricity Tariff: ₹5 per unit

Formula:

Monthly Units = Monthly Electricity Bill ÷ ₹5

Step 2 – Calculate Daily Consumption

Daily Units = Monthly Units ÷ 30

Step 3 – AI-Powered System Size Recommendation

The calculator should recommend the most suitable system size based on:

- Monthly electricity consumption
- Available roof area
- Daily generation of **4 units per kW**
- Future expansion allowance (optional 10–20%)
- Practical installation sizes (1 kW, 2 kW, 3 kW, 4 kW, 5 kW, etc.)

Formula:

Required kW = Daily Units ÷ 4

Always round up to the next whole kW.

Example:

- 2.2 kW → Recommend 3 kW
- 4.4 kW → Recommend 5 kW

If roof space is insufficient, recommend the maximum installable capacity and display:

"Based on your available roof area, we recommend installing a maximum of X kW. To achieve your ideal system size, an additional Y sq. ft. of shadow-free roof space is required."

Step 4 – Roof Area Validation

Required Roof Area = Recommended kW × 100 sq. ft.

Step 5 – Project Cost

Project Cost = Recommended kW × ₹70,000

Step 6 – Government Subsidy

System Size	Central Subsidy	Odisha State Subsidy	Total Subsidy
1 kW	₹30,000	₹25,000	₹55,000
2 kW	₹60,000	₹50,000	₹1,10,000
3 kW & Above	₹78,000	₹60,000	₹1,38,000

Apply the maximum subsidy of **₹1,38,000** for systems of **3 kW and above**.

Step 7 – Customer Investment

Customer Investment = Project Cost – Total Subsidy

Step 8 – EMI Calculator

Applicable only for Bank Loan.

Interest Rate

6.5% per annum

Use the standard EMI formula:

$$\text{EMI} = P \times r \times (1+r)^n \div ((1+r)^n - 1)$$

Display:

- Monthly EMI
 - Total Interest
 - Total Repayment Amount
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Step 9 – Monthly Solar Generation

$$\text{Monthly Generation} = \text{kW} \times 4 \times 30$$

Step 10 – Monthly Savings

$$\text{Monthly Savings} = \text{Monthly Generation} \times ₹5$$

Step 11 – Annual Savings

$$\text{Annual Savings} = \text{Monthly Savings} \times 12$$

Step 12 – 25-Year Savings Projection

Calculate the total financial benefit over the expected life of the solar system.

Formula

$$\text{25-Year Savings} = \text{Annual Savings} \times 25$$

Display:

- Total Electricity Generated (25 Years)
- Total Electricity Bill Saved
- Total Customer Investment
- Net Financial Benefit

Example

Annual Savings = ₹36,000

25-Year Savings

= ₹36,000 × 25

= ₹9,00,000

(Future enhancement: include electricity tariff escalation of 3–5% per year to provide a more realistic projection.)

Step 13 – Carbon Emission Reduction Calculator

Estimate the environmental impact of installing rooftop solar.

Assumption

1 Solar Unit Generated = 0.82 kg CO₂ Avoided

Calculations

Monthly CO₂ Reduction

= Monthly Generation × 0.82

Annual CO₂ Reduction

= Monthly CO₂ Reduction × 12

25-Year CO₂ Reduction

= Annual CO₂ Reduction × 25

Display:

- Monthly CO₂ Reduction (kg)
- Annual CO₂ Reduction (kg)
- Lifetime CO₂ Reduction (kg/tonnes)

Example

600 Units/Month

Monthly CO₂ Reduction

= 600 × 0.82

= 492 kg CO₂

Annual Reduction

= 5.9 Tonnes

25-Year Reduction

≈ 148 Tonnes CO₂

Display an easy-to-understand message:

"By installing this solar system, you can reduce approximately 148 tonnes of CO₂ emissions over 25 years, contributing to a cleaner and greener environment."

Final Results Dashboard

Display:

- Recommended Solar Capacity
 - Roof Area Required
 - Monthly Electricity Consumption
 - Monthly Solar Generation
 - Estimated Project Cost
 - Central Subsidy
 - Odisha State Subsidy
 - Total Subsidy
 - Customer Investment
 - Monthly EMI (if applicable)
 - Monthly Savings
 - Annual Savings
 - 25-Year Savings
 - Payback Period
 - Monthly CO₂ Reduction
 - Annual CO₂ Reduction
 - 25-Year CO₂ Reduction
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PDF Report Generation

Allow users to generate a professional branded PDF report after entering:

- Name
- Mobile Number

- Email Address
- District

The PDF should include:

Cover Page

- Clans Machina Logo
- Customer Name
- Date
- Proposal Number (Auto Generated)

Solar Assessment Summary

- Customer Details
- Electricity Bill
- Roof Area
- Recommended System Size

Financial Summary

- Project Cost
- Government Subsidy
- Customer Investment
- EMI
- Monthly Savings
- Annual Savings
- 25-Year Savings
- Payback Period

Environmental Impact

- Monthly CO₂ Reduction
- Annual CO₂ Reduction
- 25-Year CO₂ Reduction

Why Choose Clans Machina

- Premium Solar Solutions
- MNRE-Compliant Installation
- Government Subsidy Assistance
- Bank Loan Support
- Net Metering Assistance
- Comprehensive Warranty
- Professional Installation
- Dedicated After-Sales Service

Call to Action

- Book a Free Site Survey
 - Contact Solar Expert
 - Scan QR Code (optional)
 - Customer Care Number
 - Website URL
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Lead Capture & CRM Integration

Before allowing the PDF download or detailed proposal request, capture:

- Name
- Mobile Number
- Email Address
- District
- Monthly Electricity Bill
- Roof Area
- Recommended System Size

Automatically send the lead to:

- CRM
- Sales Team
- WhatsApp
- Email Notification

Future Enhancements

- Google Maps roof measurement
- Satellite-based roof estimation
- DISCOM-wise tariff selection
- Battery backup recommendation
- EV charging compatibility recommendation
- Live subsidy updates
- AI chatbot assistance
- WhatsApp estimate sharing
- Online site survey booking
- Proposal approval workflow
- Digital payment integration
- Real-time project tracking after order confirmation

